

CALIFORNIA WILDFIRE OCCURRENCE MODELLING

Methodological Development, Diagnostic Audit, Feature Selection, and Preprocessing Screening

Comprehensive research report through 4 August 2026

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Audience: thesis/paper supervision and technical machine-learning review

Status: working methodology report; final model selection and test evaluation remain pending

The period 29 July-4 August 2026 is treated as the central methodological contribution for the planned paper.

Document control and current status

Item	Current status
Prediction unit	California EPA Level III ecoregion-day
Target	Binary indicator: at least one recorded wildfire occurred in the ecoregion on the day
Main development split	2013-2020 training; 2021-2022 validation; 2023-2025 out-of-time test
Primary model family	LightGBM classifier; XGBoost planned as a separately frozen comparison
Primary selection metric	Average precision (AP); Brier score and log loss as probability-quality metrics
Frozen feature candidate	36-feature compact set excluding transmission and road/population infrastructure variables
Current preprocessing finalists	ExtraTrees iterative + SMOTENC; native LightGBM + no resampling; median + SMOTENC; RF iterative + none (conditional)
Final test status	Not yet run under the final frozen pipeline; descriptive test properties were previously inspected

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Executive summary

Central conclusion

The work completed from 29 July to 4 August converted an exploratory modelling pipeline into a defensible, leakage-aware and temporally validated methodology. The strongest current feature specification is the 36-feature compact model without transmission density. The strongest screening pipeline for mean ranking performance is ExtraTrees iterative imputation followed by SMOTENC, while native LightGBM without resampling remains the essential simplicity and robustness control. No final pipeline should be declared until duplicate configuration rows are removed, missing/failed experiments are resolved, finalist out-of-fold predictions are compared by paired temporal block bootstrap, and calibration is evaluated on the fixed 2021-2022 validation period.

- The latest documented dataset version contains 1,331 training fires, 284 validation fires and 820 test fires, with fire rates of approximately 3.5%, 3.0% and 6.5%, respectively. Earlier logs contain slightly different counts, demonstrating that dataset versioning must be made explicit.
- The audit found no duplicate rows or duplicate ecoregion-date records, but identified major temporal drift in year and several infrastructure variables, extensive redundancy among weather summaries and missingness indicators, and approximately 40% NDVI missingness in the test period.
- A fixed moderate-capacity LightGBM was used as a measuring instrument for feature-family and within-family ablations. This separated feature selection from later hyperparameter optimisation.
- The 36-feature compact model without transmission achieved the strongest mean results across four rolling chronological folds: mean AP about 0.253, mean AP/prevalence about 6.93, mean Brier about 0.0333 and mean log loss about 0.1567.
- Paired temporal block bootstrap supported the compact model over the full baseline for AP and log loss. Removing transmission clearly improved Brier and log loss relative to the transmission-inclusive compact model, while the AP interval narrowly crossed zero.
- The absence of explicit human-activity variables is a genuine limitation, but weak or unstable infrastructure proxies should not be forced into the primary model merely to satisfy a conceptual expectation. The model should be described as weather-conditioned wildfire occurrence susceptibility rather than a complete causal model of ignition.
- The categorical ecoregion feature invalidated ordinary SMOTE. The corrected pipeline imputes and encodes ecoregion within each training fold and uses SMOTENC, preventing synthetic fractional category codes and leakage.
- The sampler screen showed a small mean AP advantage for SMOTENC but a temporal reversal: sampling helped earlier folds and hurt later folds. Native LightGBM therefore remains a serious finalist.
- The imputer screen placed ExtraTrees iterative + SMOTENC first on mean AP, but its advantage over median + SMOTENC was small relative to runtime. Random Forest iterative + no sampling was scientifically competitive but operationally expensive and showed a suspicious Fold 4 runtime anomaly.

Current decision summary

Decision	Status	Rationale
Primary feature set	Select	36-feature compact set without transmission; best mean rolling-fold performance and supported probability-quality improvements.
Transmission feature	Sensitivity only	Useful in one fixed validation period, but less consistent across historical folds and worse Brier/log loss in paired bootstrap.
Road/population variables	Do not advance	High drift, coarse stepwise distributions and no consistent gain over the compact set.
Native LightGBM + none	Mandatory finalist	Simple, fast, strong Brier score and better AP in the two latest rolling folds.

Decision	Status	Rationale
ExtraTrees iterative + SMOTENC	Performance finalist	Highest mean AP and strongest SMOTENC imputer, but more expensive and temporally regime-dependent.
Median + SMOTENC	Efficiency finalist	Near-leading AP at seconds rather than tens of minutes.
RF iterative + none	Conditional finalist	Consistent AP gains versus native, but tiny effect size, extreme runtime and one anomalous timing result.
Class weighting / random oversampling	Reject	Inferior mean AP and/or substantially poorer uncalibrated probability metrics.

1. Research objective and modelling scope

The long-term project objective is to estimate daily wildfire occurrence probabilities for California and, eventually, for user-selected U.S. states. The operational unit is an EPA Level III ecoregion-day. A positive target means that one or more recorded fires occurred in that ecoregion on that date. The envisioned application combines recent or forecast weather with stable geographic information and returns daily probabilities for the next several days.

The current model is not a fire-spread model, burned-area model or causal ignition-source model. It predicts observed binary occurrence. This distinction became important when the infrastructure variables were removed: the remaining model primarily represents environmental receptiveness and weather-conditioned susceptibility, although ecoregion and seasonality may indirectly absorb stable human patterns.

Recommended interpretation of the output

Daily probability of recorded wildfire occurrence conditional on forecastable weather, vegetation, ecoregion and data-quality indicators. Avoid describing the output as the probability that humans will start a fire or as a complete causal ignition model.

1.1 Core statistical challenges

- Rare-event classification: fire prevalence is usually between about 1% and 5% in rolling development folds, with substantial variation between periods.
- Temporal nonstationarity: annual fire rates, weather relationships, missingness and spatial exposure change over time.
- Spatial dependence: multiple ecoregions are observed on the same date, and observations within ecoregions are repeated through time.
- Feature availability: remote-sensing and infrastructure variables may be incomplete or stale at forecast time.
- Probability quality: the final product requires meaningful probabilities, not only binary rankings.
- Anthropogenic omission: the target combines human- and lightning-caused fires while the strongest available predictors are mostly environmental.

2. Dataset architecture and split definition

2.1 Data sources and feature families

The assembled dataset spans 2013-2025 and combines wildfire incident labels with NOAA/NCEI weather observations aggregated from hourly station records to daily ecoregion-level summaries. Additional variables represent vegetation, time, ecoregion identity, infrastructure and observation completeness. The latest feature inventory before reduction contained approximately 62-67 candidate predictors, depending on dataset and experiment version.

Family	Representative variables	Research role
Temperature	mean, minimum, maximum, standard deviation, DTR	Ambient heat and diurnal variability.
Moisture / atmospheric dryness	dew point, RH, saturation vapour pressure, VPD	Fuel drying and atmospheric demand.
Wind	speed summaries, east-west and north-south components, HDWI proxy	Ignition receptiveness and rapid growth conditions.
Visibility / pressure / ceiling	visibility, sea-level pressure, ceiling height	Synoptic and observation-state indicators.
Vegetation	NDVI and EVI summaries	Fuel amount and greenness.
Spatial / infrastructure	population density, road density, transmission density and derived risk	Potential anthropogenic exposure and stable geographic structure.
Temporal / categorical	month, year, US_L3NAME	Seasonality, trend and ecoregion differences.
Data completeness	counts, missing rates and missing-flag counts	Observation coverage and possible station-network effects.

2.2 Chronological split

Partition	Years	Purpose	Latest fires	Latest rate
Training	2013-2020	Feature selection, rolling folds and pipeline development	1,331	0.035
Validation	2021-2022	Final calibration, threshold and pipeline selection after development freeze	284	0.030
Test	2023-2025	One final out-of-time evaluation after complete freeze	820	0.065

The latest data log states that there are no duplicate rows. Earlier July logs reported 1,321/260/817 fires and slightly different row counts. The later log reports 1,331/284/820. The change is not fully explained in the available notes and must be tied to a version-controlled dataset rebuild or correction before publication.

Test-set transparency

The 2023-2025 period has not been used for final model scores in the new screening methodology, but its prevalence, missingness and feature distributions were inspected. It is therefore an out-of-time test set, but not a perfectly blinded holdout. The paper should disclose this and avoid any further test-driven decisions.

2.3 Target imbalance across time and geography

Fire occurrence is strongly seasonal and geographically concentrated. In the latest log, summer has a fire rate near 9.8%, compared with approximately 0.5% in winter. July, June and August have the highest monthly rates. The Central California Foothills and Coastal Mountains ecoregion has a substantially higher rate than the northern basin and coastal ecoregions. Annual rates also vary sharply, with 2024-2025 much higher than 2023 and many earlier years.

These patterns justify chronological evaluation and retention of seasonality and ecoregion information. They also explain why a single random split or one aggregate performance number would be misleading.

3. Foundation before 29 July 2026

Before the focused audit, the project had already progressed through dataset construction, weather parsing, feature engineering and several model families. Important lessons from this earlier phase motivated the more rigorous July-August methodology.

- A dense neural-network baseline under severe imbalance predicted almost all zeros and produced ROC-AUC near chance despite high accuracy. SMOTE and focal-loss variants improved recall but retained low PR-AUC, illustrating why accuracy was unsuitable.
- LightGBM became the main model because it handled nonlinear interactions, large mixed feature families and missing values efficiently.
- An earlier direct comparison found that LightGBM with native missing values outperformed imputation-only and imputation-plus-scaling variants on an earlier dataset. This made native handling an essential control in all later screens.
- Initial imputation studies included median, Bayesian iterative, KNN, Random Forest, ExtraTrees, MissForest and SoftImpute variants, together with multiple imbalance methods. Early two-iteration searches were treated as smoke tests rather than substantive evidence.
- A later 45-iteration run suggested MissForest, ExtraTrees-regressor imputation, simple imputation and a standard preprocessing condition as candidates, but the experiment combined many moving parts and did not yet include the systematic feature audit developed later.
- The initial scripts calibrated every candidate and evaluated the test period during development. The revised workflow separates feature selection, preprocessing screening, tuning, calibration and one final test.

The main methodological weakness before 29 July was not insufficient modelling sophistication. It was the absence of a frozen, audited feature set and a clean hierarchy of questions. The July-August work corrected this by making each experimental stage answer one question while holding the remaining choices constant.

4. Detailed chronology of the new methodology: 29 July-4 August 2026

Chronology of the methodological development

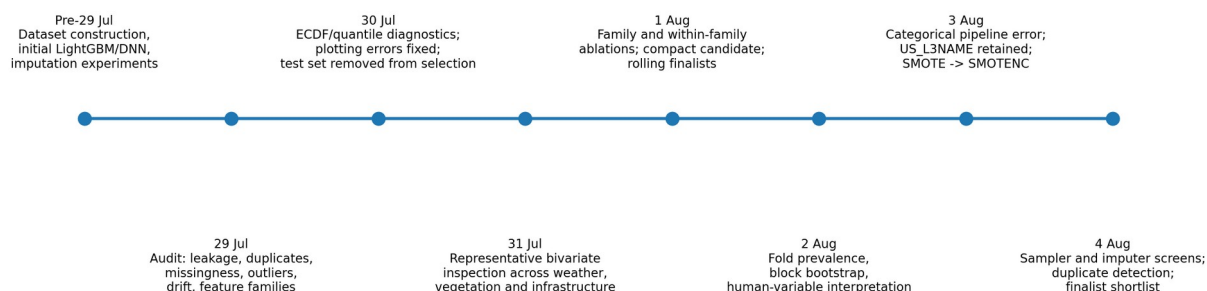


Figure 1. Chronology of the methodological development. The period 29 July-4 August is the main contribution of this report.

4.1 29 July: dataset audit and experiment hierarchy

- The prediction unit, chronological split and forecast-time availability of candidate predictors were re-examined.
- Duplicate rows and duplicate ecoregion-date records were checked and none were found.
- Data types were reviewed: date needed datetime handling and ecoregion identifiers required categorical treatment rather than accidental continuous encoding.
- Missingness, outliers, fire rates by year/month/season/ecoregion, descriptive split statistics and PSI were assembled.
- Correlation was reframed as a redundancy diagnostic rather than an automatic deletion rule. For a tree model, high Spearman correlation alone was judged insufficient for removal.
- A staged workflow was established: audit -> family ablation -> compact feature freeze -> sampler screen -> imputer screen -> finalist matrix -> tuning -> calibration/threshold -> final test.

An initial note stated that both EVI and NDVI were unavailable through the test period. Later inspection corrected this: EVI was complete, whereas NDVI had approximately 40% missingness in the test period. The correction materially affected the vegetation decision.

4.2 30 July: bivariate diagnostic design and plotting corrections

The diagnostic plot combined a fire-event-only ECDF with observed fire rates in training-derived quantile bins. The training distribution defined the bin edges, and validation observations were mapped into those fixed bins. Wilson intervals were used as descriptive uncertainty bars.

- A `TypeError (unhashable DataFrame)` occurred because the test `DataFrame` was passed positionally into the target argument. The call was corrected to use training and validation only.
- A second issue came from applying `pd.qcut` to a `DataFrame` rather than a `Series`. The feature `Series` was used explicitly.
- Validation values outside the training range produced missing bins. The first and last bin edges were extended to negative and positive infinity.
- The test period was removed from routine feature-selection plots.
- The no-fire rate plot was judged redundant because it equals one minus the fire rate. A no-fire ECDF could be used only as a separate drift diagnostic.
- Shaded confidence bands were removed when error bars already communicated the interval.

4.3 31 July: representative ECDF and quantile inspection

Representative variables were inspected across vegetation, weather and infrastructure. The purpose was not to select a feature solely from a graph, but to identify stable nonlinear relationships, drift, coarse spatial steps and interactions worth testing by ablation.

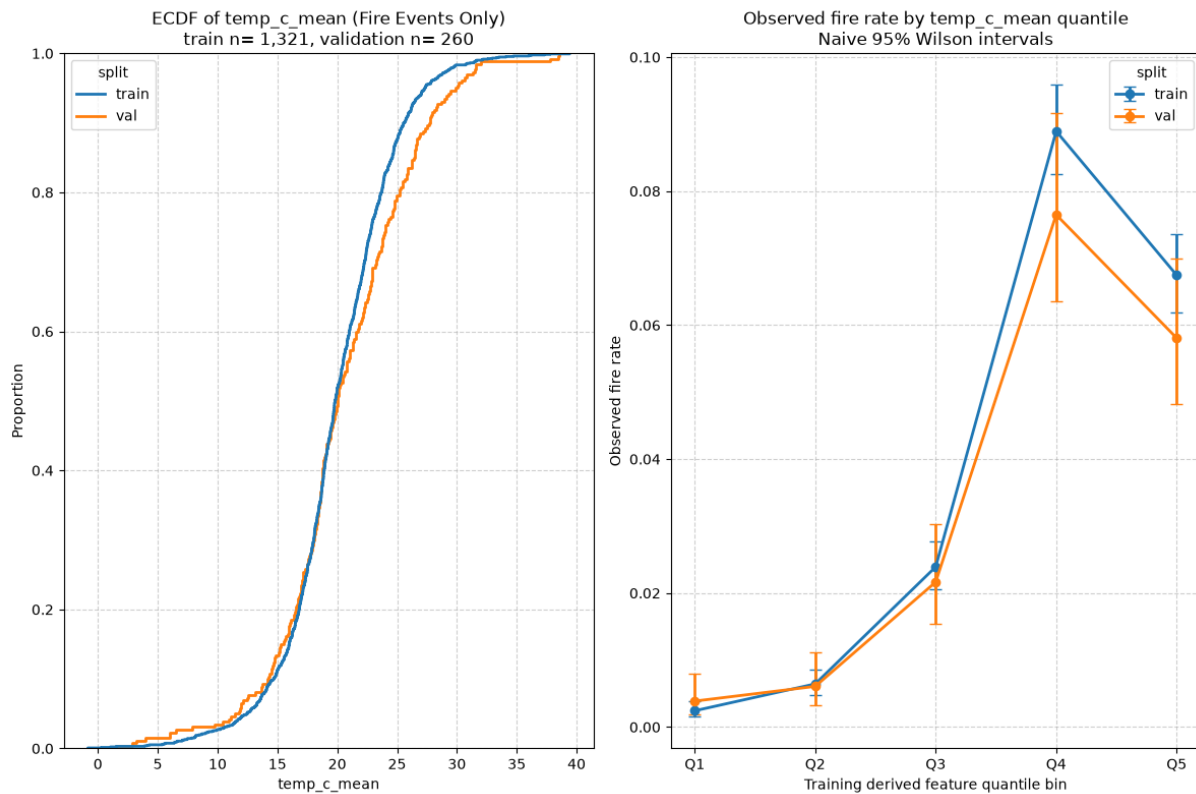


Figure 2. Example diagnostic for mean temperature. The fire rate rises strongly through the fourth training-derived quantile and declines in the fifth; the broad shape is preserved in validation. Wilson intervals are descriptive and do not account for temporal or spatial clustering.

Feature	Observed diagnostic	Implication
temp_c_mean	Validation fire events were somewhat warmer, but the quantile-risk shape was stable.	Retain a temperature representation; do not rely on one summary without ablation.
HDWI_proxy	Validation positive events shifted higher; both splits rose through Q4 and declined in Q5.	Physically meaningful but not sufficient as the only wind/dryness feature.
evi_mean	Moderate positive-event shift; similar middle-quantile shape but weaker Q5 validation rate.	EVI remained operationally available and worth retaining.
ndvi_mean	Positive-event distribution shifted and the middle quantile was unstable; future missingness was severe.	Exclude from primary operational set.
road density	Stepwise spatial distribution and unstable Q4/Q5 rates.	Potential proxy with high drift; avoid in conservative primary model.
transmission density	Coarse, stable spatial steps and broadly similar validation shape.	Most defensible infrastructure sensitivity variable, but still static and correlated.
visibility / pressure	Relatively stable ECDFs and nonlinear quantile relationships.	Retain compact representatives if ablation supports them.

4.4 1 August: family ablations and compact finalists

A fixed moderate-capacity LightGBM was used to isolate feature inclusion effects. Hyperparameters, imbalance treatment and all other modelling choices were held constant. The full baseline was compared with family removals, within-family reductions and selected combinations.

```
LGBMClassifier(
    objective='binary', n_estimators=300, learning_rate=0.05,
    num_leaves=31, max_depth=8, min_child_samples=50,
    subsample=1.0, colsample_bytree=1.0,
    class_weight=None, random_state=42
)
```

The compact 37-feature candidate achieved the strongest fixed 2021-2022 validation AP (0.2158 versus 0.1865 for the 62-feature baseline). A 36-feature version without transmission obtained AP 0.2031. Because the fixed validation result could be period-specific, these candidates were moved to rolling chronological folds rather than selected from the single split.

Experiment	Features	AP	Brier	Log loss	Interpretation
Compact candidate	37	0.2158	0.02706	0.11213	Highest fixed validation AP; includes transmission.
Spatial keep road	58	0.2036	0.02711	0.11030	Good threshold metrics, but road drift was high.
Compact without transmission	36	0.2031	0.02712	0.11204	Strong compact alternative and more conservative spatial design.
EVI standard deviation only	55	0.2026	0.02779	0.11206	Vegetation family contained useful signal but also redundancy.
EVI mean and standard deviation	56	0.2008	0.02739	0.10984	Strong log loss; supported compact EVI representation.
Baseline all features	62	0.1865	0.02723	0.11142	Reference model; manually inserted in the logged table and later required reproducibility audit.

4.5 2 August: prevalence-normalised rolling folds and paired bootstrap

Four rolling chronological folds were formed from unique dates in 2013-2020, preventing rows from the same calendar date from appearing on both sides. Every validation fold contained 7,592 rows, but fire prevalence ranged from 1.30% to 5.37%. AP was therefore reported together with AP/prevalence.

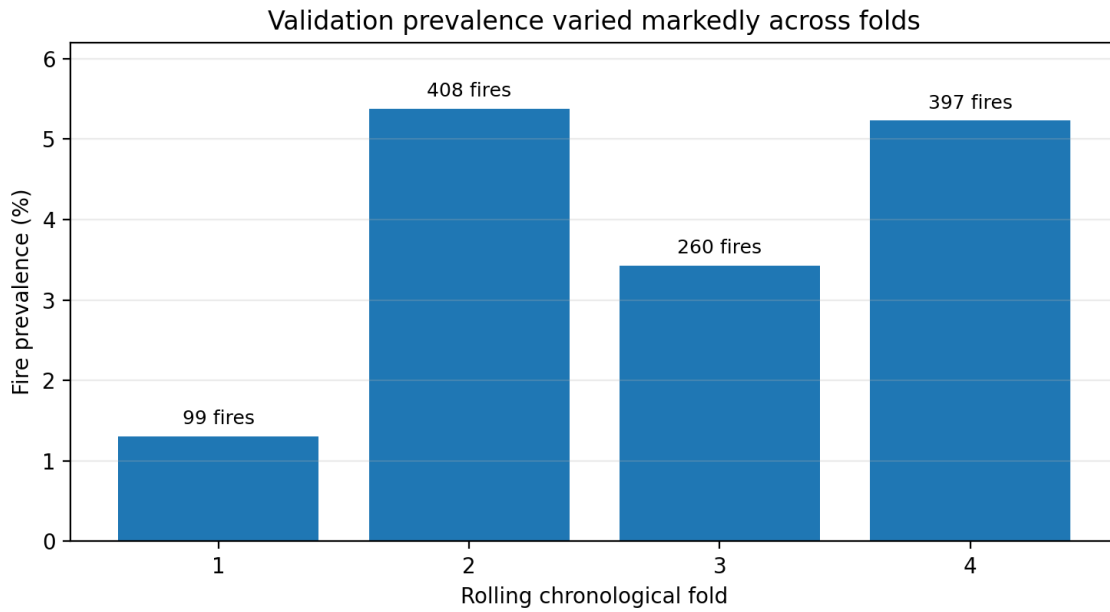


Figure 3. Fire prevalence differed substantially across the rolling validation periods. AP/prevalence was used as context, not as a replacement for AP.

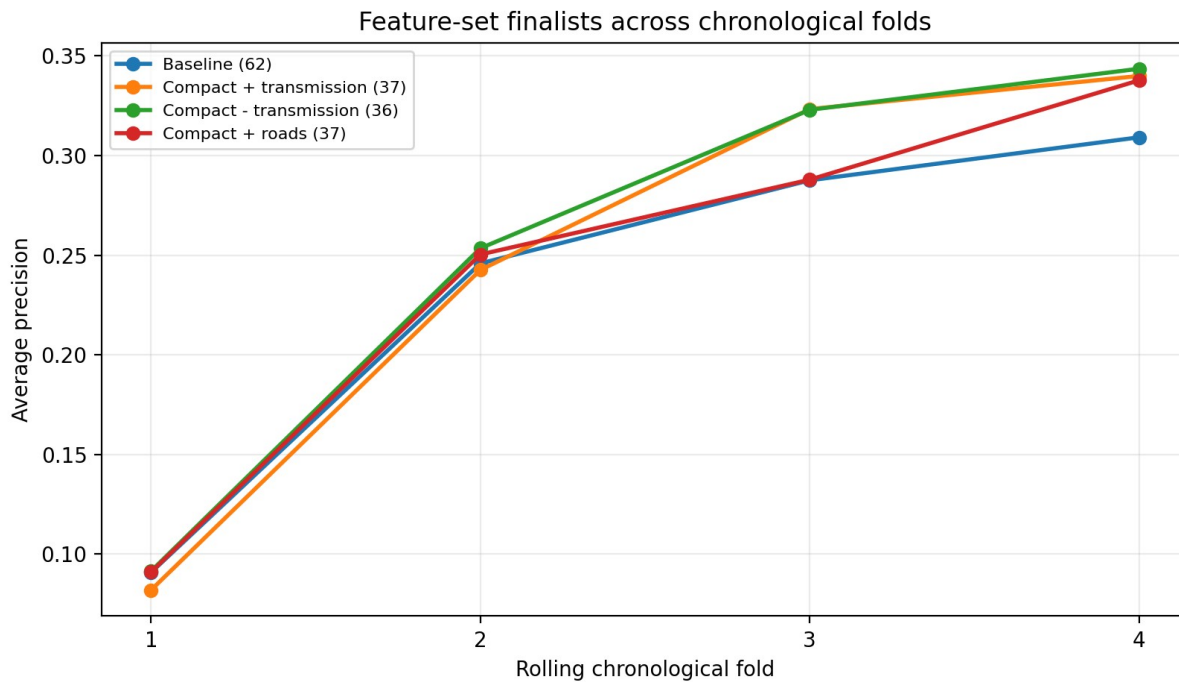


Figure 4. The 36-feature compact model without transmission beat the 62-feature baseline in all four folds and was the strongest mean finalist.

The compact model without transmission achieved mean AP 0.2529, mean AP/prevalence 6.93, mean Brier 0.03327 and mean log loss 0.15672. It beat the baseline in AP and Brier in every fold. The transmission-inclusive compact model was slightly weaker on average and less consistent.

Feature set	Mean AP	Mean AP/prevalence	Mean Brier	Mean log loss
Baseline all features	0.2334	6.46	0.03369	0.16135
Compact + transmission	0.2470	6.69	0.03341	0.15879

Feature set	Mean AP	Mean AP/prevalence	Mean Brier	Mean log loss
Compact - transmission	0.2529	6.93	0.03327	0.15672
Large road-based set	0.2400	6.63	0.03354	0.16057
Compact + roads	0.2418	6.63	0.03353	0.15913

4.6 2 August: anthropogenic-variable concern

The removal of transmission, road and population variables raised a legitimate scientific concern: most U.S. and California wildfires are human-started in many datasets. The methodological response was not to force weak proxies into the model. Instead, the primary model was reinterpreted as weather-conditioned occurrence susceptibility, transmission was retained as a sensitivity model, and richer anthropogenic variables were proposed for future work.

4.7 3 August: categorical-data failure and SMOTENC correction

The first screening script stopped with: 'This screening script currently assumes numeric predictors. Non-numeric columns found: [US_L3NAME].' This was a valuable failure because it exposed a methodological incompatibility rather than a superficial type error.

Ordinary SMOTE cannot safely operate on arbitrary integer codes for nominal ecoregions. It could interpolate category codes and create synthetic values that do not correspond to any ecoregion. The corrected pipeline separated numeric and categorical processing and replaced ordinary SMOTE with SMOTENC.

```

Training fold only
  numeric predictors -> fold-fitted imputer -> scale when required
  US_L3NAME -> most-frequent imputation -> fold-fitted ordinal encoder
  combined matrix -> SMOTENC or no sampler
  LightGBM(categorical_feature=[ecoregion_index])
Validation fold remains untouched

```

4.8 4 August: sampler and imputer screens

The corrected screening script deliberately excluded calibration, threshold selection and the 2023-2025 test period. It first compared imbalance methods under median imputation and native handling, then compared imputers under no sampling and SMOTENC.

- Sampler screen: median + SMOTENC had the highest mean AP, but its advantage over native handling was only about 0.0023 and reversed in the two latest folds.
- Class weighting substantially worsened Brier and log loss and was removed.
- SMOTENC-Tomek did not outperform ordinary SMOTENC overall and was deferred.
- Imputer screen: ExtraTrees iterative + SMOTENC ranked first by mean AP; RF iterative + no sampling ranked second; native handling remained highly competitive.
- The fold-results CSV contained duplicated native+none rows, which preserved means but distorted standard deviations, runtimes and paired comparisons.
- MissForest results were absent, indicating failure or exclusion that must be documented rather than silently ignored.

5. Data audit: rationale, methods and findings

5.1 Leakage and forecast-time availability

Every candidate predictor was assessed against the intended deployment time. The target `fire_occurred` and the auxiliary `fire_count` must never enter `X`. Raw date is retained for splitting and reporting, not modelling. Year is known at prediction time but has severe monotonic drift and poor tree extrapolation properties, so it was removed from the primary set. Ecoregion identity is stable and available. EVI is complete in the latest data; NDVI is incomplete in the future period.

- Direct target leakage removed: `fire_count` and `fire_occurred` excluded from predictors.
- Identifier redundancy reduced: use `US_L3NAME` as the categorical ecoregion representation rather than both code and name.
- Raw date excluded after deriving month/seasonality; year retained only for splitting/reporting and ablation.
- Observation-count and missingness features treated as possible station-network proxies, not assumed to be neutral meteorology.
- Infrastructure variables required explicit confirmation that their historical values were available at the intended prediction date.

5.2 Missingness and observation quality

Missingness was not treated merely as a preprocessing inconvenience. It was evaluated as a source of drift and potential predictive signal. Weather missing rates varied by variable and period. Ceiling height had the largest missingness among retained weather variables; sea-level pressure was also materially incomplete. Several missing-flag count columns were exact duplicates and therefore redundant.

Derived completeness feature

`row_weather_missing_rate` is the row-wise mean of eight weather missing-rate columns. It provides a compact observation-quality summary while selected variable-specific missing rates are retained where they carry distinct information. The mean is defensible only because the rates are on the same 0-1 scale; weighted aggregation would be preferable if denominators differ materially.

5.3 Outliers

IQR-based outlier percentages were recorded as diagnostic information, not as an automatic deletion rule. Visibility, infrastructure densities, wind minima, HDWI and several missingness rates showed many flagged values. For environmental variables, extreme observations can be physically meaningful and useful to tree models. The audit therefore focused on impossible values, unit errors and documentation rather than blanket winsorisation.

5.4 Distribution drift

Feature or family	Key finding	Decision consequence
year	PSI approximately 7.3; deterministic separation by split.	Remove from primary predictive set; retain for split/reporting.
road density	Large PSI in different dataset versions and coarse ecoregion-level steps.	Do not use in conservative primary model.
population density	Large train-test drift in one audit and high correlation with road density.	Do not advance as primary anthropogenic proxy.

Feature or family	Key finding	Decision consequence
NDVI	Approximately 40% missingness in test; notable distribution changes.	Exclude from primary operational model.
EVI	Complete across latest splits; moderate but manageable drift.	Retain mean and standard deviation.
temperature, wind, pressure	Generally modest mean shifts; some positive-event drift.	Retain with chronological validation.
weather missing rates	Some PSI above 0.2 and changing missingness.	Retain compact completeness indicators and monitor operationally.

5.5 Correlation and redundancy

High correlation was used to organise variables into logical families, not to delete features mechanically. Tree models are less sensitive to multicollinearity than linear models, but redundant variables can dilute importance, increase computation, encourage arbitrary split substitution and weaken interpretability.

Redundant group	Approximate strongest correlation	Resolution
Temperature mean/min/max	0.95-0.96	Evaluate compact alternatives by ablation; final set retained several temperature summaries because joint interactions remained useful.
Temperature standard deviation vs DTR	0.98	Recognise derived redundancy; retain only if joint compact model improves temporally.
Dew-point summaries	0.97-0.98	Do not select from univariate AUC alone; test humidity/dryness representations jointly.
Pressure summaries	0.95-0.97	Compact mean representation preferred in interpretation, though final 36 list retained mean only.
Observation counts	0.97-1.00	Remove duplicated and broad count family from primary set.
Missing-flag counts	1.00 for many variables	Retain at most one representative plus rate summaries.
Road/transmission/population	about 0.9 or higher for several pairs	Avoid redundant static proxies in primary model.

6. Bivariate diagnostic methodology

6.1 Why ECDF plus training-derived quantile bins

The ECDF among fire events reveals whether the distribution of a feature conditional on a positive event shifts between training and validation. The quantile panel estimates observed fire rate across training-derived bins and tests whether the relationship shape transfers to validation. This combination is more informative than a simple correlation coefficient because it can reveal nonlinearity, thresholds and tail reversals.

The bins were defined using the training feature distribution only. Validation values were mapped to those fixed edges, with infinite outer boundaries to capture values outside the training range. This prevents validation from influencing the bin definitions.

6.2 Interpretation limits

- Wilson intervals are descriptive. Rows are not independent because dates and ecoregions are clustered.
- Fire-event ECDFs condition on a relatively small positive sample and can be noisy.
- A visually stable relationship does not establish causal importance or multivariate incremental value.
- A visually unstable feature is not automatically useless if it interacts with other predictors.
- The plots support ablation design; they do not replace rolling validation.

6.3 Selected diagnostic examples

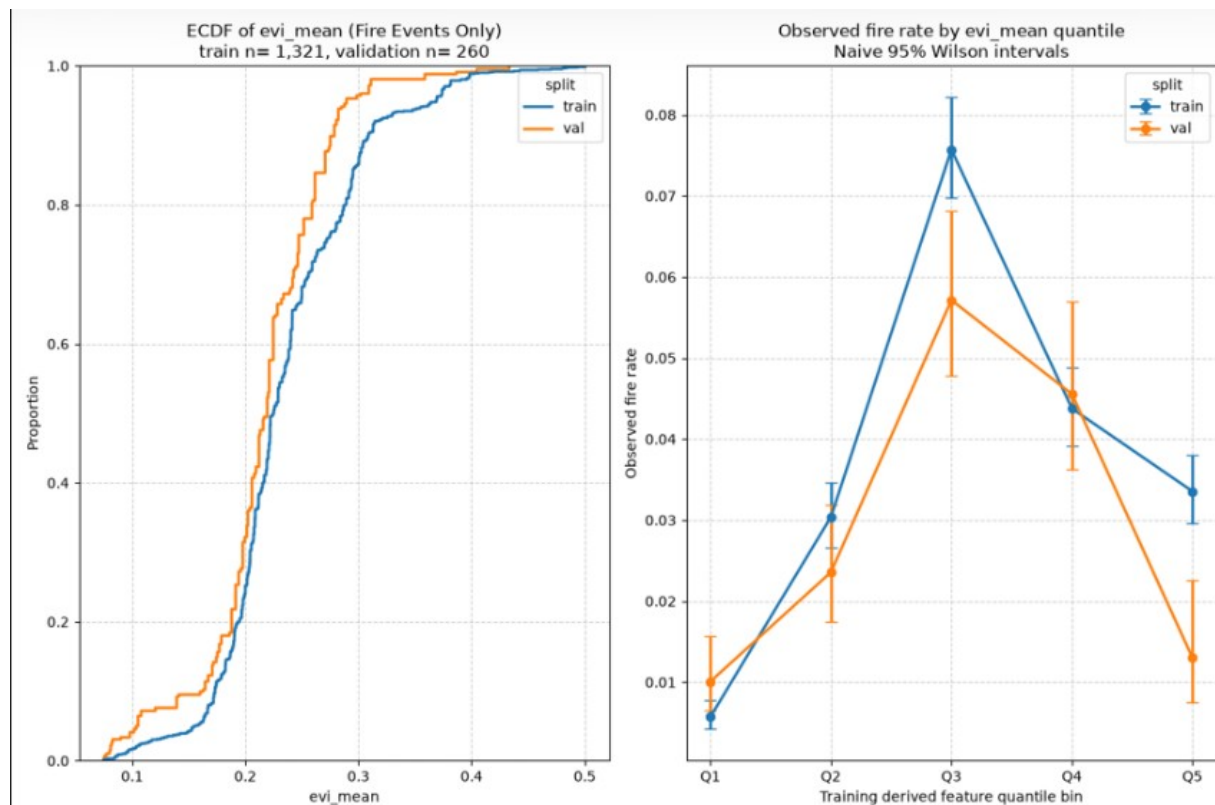


Figure 5. EVI mean diagnostic: moderate event-distribution shift, but a persistent nonlinear relationship and complete future availability supported retaining compact EVI features.

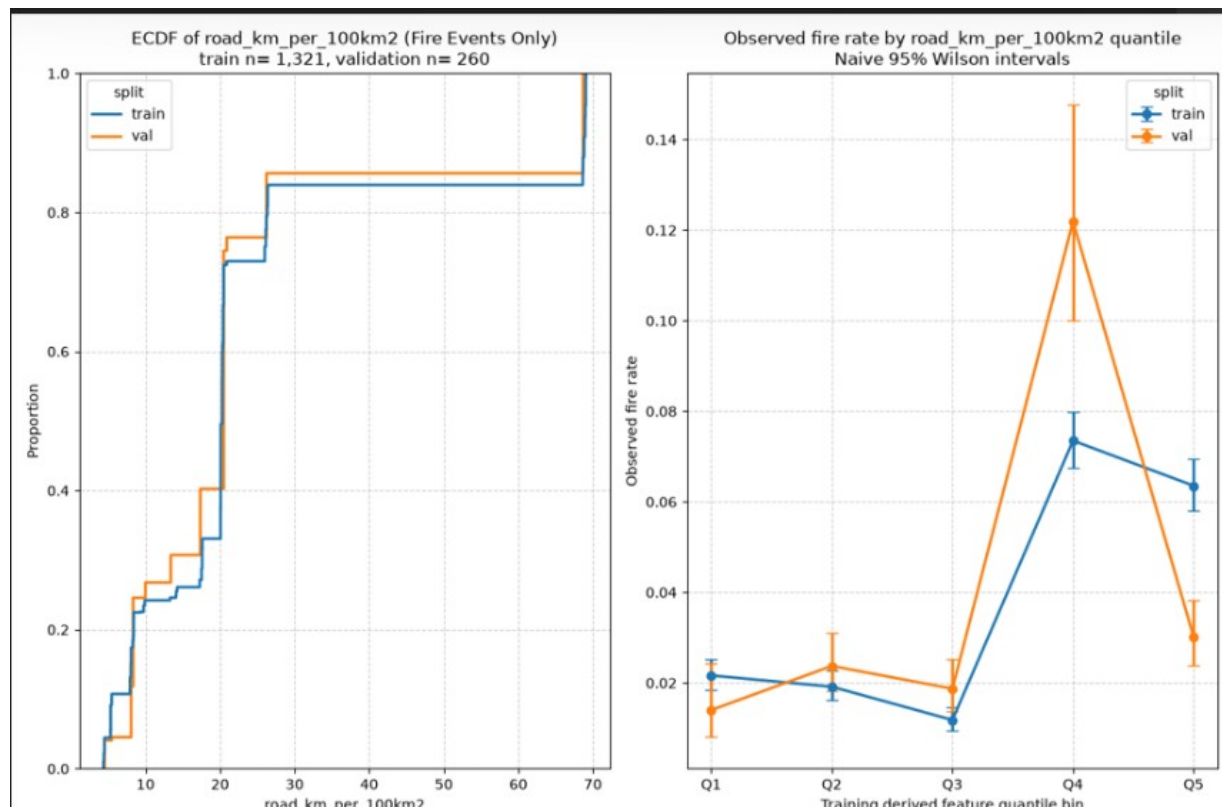


Figure 6. Road-density diagnostic: coarse spatial steps and unstable upper-quantile validation rates were consistent with proxy and drift concerns.

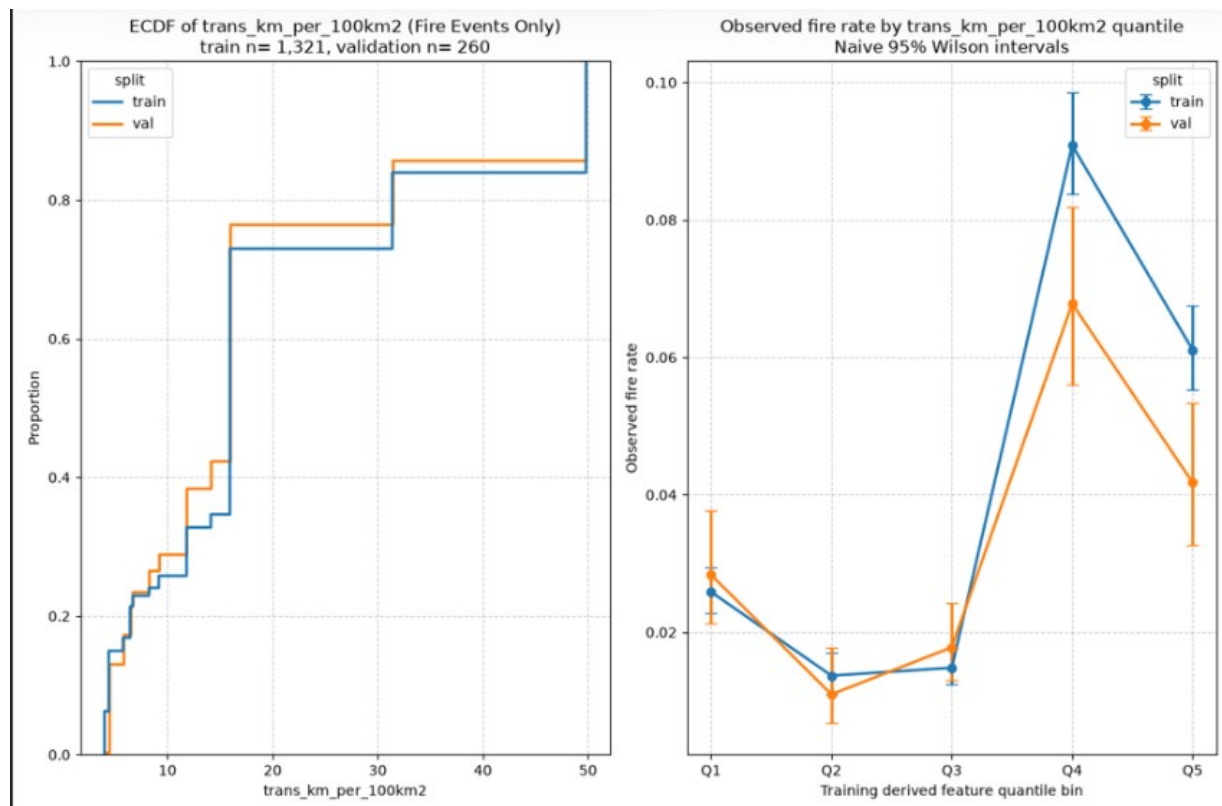


Figure 7. Transmission-density diagnostic: stable coarse spatial steps and a plausible upper-quantile signal, but limited evidence of robust incremental value after chronological testing.

7. Feature-family and within-family ablation

7.1 Experimental principle

The feature experiments were designed as controlled ablations, not hyperparameter searches. A fixed deterministic LightGBM measured the change caused by adding or removing features. Average precision was primary because of class imbalance. Brier score and log loss were secondary probability metrics. Threshold metrics at 0.1 were retained only as diagnostics and were not used to decide the feature set.

7.2 Family-level findings

Family	Evidence	Final treatment
Temperature	Strong univariate discrimination and stable nonlinear relationship; redundant summaries.	Retain compact but sufficiently rich temperature representation.
Dew point / VPD	Useful but highly redundant; different representations traded AP against log loss.	Retain selected dryness/humidity variables; evaluate as group.
Wind	Family removal harmed AP; HDWI alone was insufficient.	Retain HDWI, directional components and wind summaries in compact set.
Visibility / pressure / ceiling	Mostly modest individual effects; some compact representatives improved probability metrics.	Retain mean/max representatives and missing rates where supported.
Vegetation	Full family contained noise; compact EVI representation improved results; NDVI operationally incomplete.	Retain EVI mean and standard deviation; drop NDVI.
Observation counts	Highly correlated station-coverage proxies; removal often neutral or beneficial.	Drop counts from primary set.
Missingness	Family removal harmed some ablations; duplicate flags were redundant.	Retain compact rate indicators and one representative flag count.
Infrastructure	Some isolated AP gains but large drift and coarse spatial structure.	Exclude from primary; transmission sensitivity only.
Temporal / ecoregion	Month and ecoregion encode strong stable structure; raw year is high-drift.	Retain month and US_L3NAME; exclude year.

7.3 Baseline inconsistency and reproducibility warning

Across successive experiment files, the all-feature baseline AP changed materially (for example approximately 0.2016 in one table and 0.1865 in the later 62-feature table). One table explicitly stated that the baseline was manually added. Such a discrepancy can arise from a changed dataset, feature list, preprocessing path, seed or row set. Until all candidates, including the baseline, are generated through the same function and versioned data, only within-table comparisons are valid.

Required reproducibility control

Every experiment must record dataset hash/version, row counts, positive counts, exact feature list, categorical handling, model parameters, random seed, fold indices and software versions. The baseline must be executed through the same code path as every candidate.

8. Compact feature selection and uncertainty analysis

8.1 Frozen 36-feature set

The primary compact set excludes transmission, roads, population, NDVI, observation counts, raw year and most exact duplicate missingness flags. It retains ecoregion, month, EVI, physically interpretable weather summaries and selected data-completeness indicators. The exact feature list is provided in Appendix A.

8.2 Rolling-fold result

The compact model without transmission improved AP over the baseline in all four rolling folds. Because validation prevalence varied, AP/prevalence was also reported. The compact model had the best mean AP, mean AP/prevalence, Brier and log loss among the feature-set finalists.

8.3 Paired temporal block bootstrap

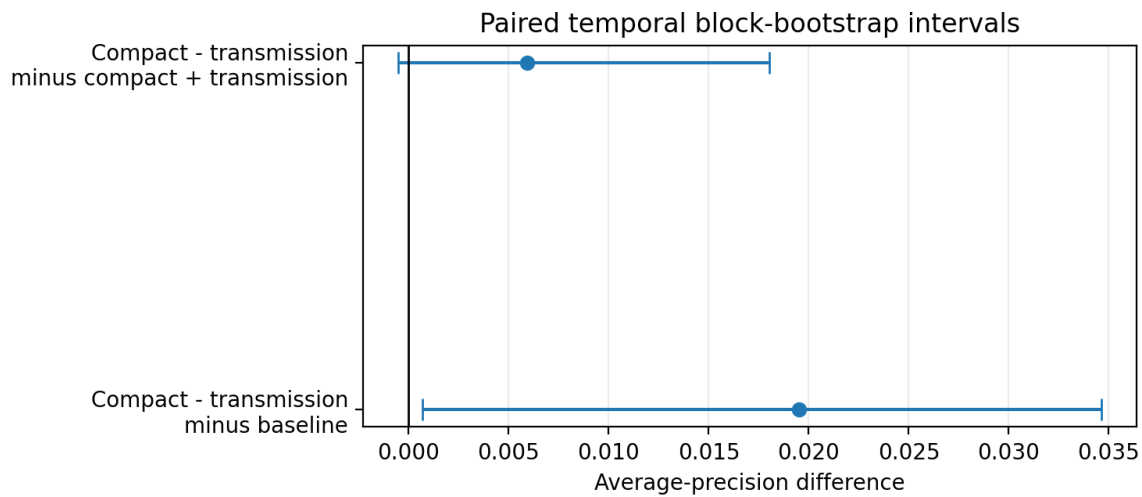


Figure 8. Paired temporal block-bootstrap intervals for AP differences. The compact model is supported over the baseline; the AP advantage over the transmission-inclusive compact model is probable but not conclusive at the 95% level.

Comparison	Metric	Point estimate	Bootstrap SE	95% interval	Pr(favourable)
No transmission - baseline	AP difference	+0.01951	0.00858	[+0.00070, +0.03463]	0.9785
No transmission - baseline	Brier difference	-0.00042	0.00027	[-0.00094, +0.00010]	0.9310 below zero
No transmission - baseline	Log-loss difference	-0.00462	0.00208	[-0.00864, -0.00062]	0.9885 below zero
No transmission - with transmission	AP difference	+0.00591	0.00466	[-0.00051, +0.01803]	0.9640
No transmission - with transmission	Brier difference	-0.00014	0.00007	[-0.00030, -0.00001]	0.9835 below zero
No transmission - with transmission	Log-loss difference	-0.00207	0.00056	[-0.00326, -0.00108]	1.0000 below zero

Interpretation: the primary compact model has supported gains over the baseline in ranking and log loss. Relative to the transmission-inclusive compact model, probability quality clearly improves when transmission is removed, while the AP advantage remains uncertain because the interval narrowly includes zero.

9. Anthropogenic-variable concern and scientific interpretation

9.1 Why the concern is valid

Human ignitions dominate many U.S. and California fire records. Balch et al. (2017) reported that human-started fires accounted for 84% of U.S. wildfires in their study period and expanded the spatial and seasonal fire niche. Mann et al. (2016) argued that anthropogenic variables contributed substantial explanatory power in California fire-probability models. Keeley and Syphard (2018) documented long-term changes in California ignition sources, and Chen and Jin (2022) found that roads, settlement and nighttime light shaped human-caused ignition probability while lightning ignitions followed different controls.

9.2 Why weak proxies should not be forced into the primary model

Transmission density represents only a narrow subset of human ignition mechanisms. Road density and population are static, coarse and correlated with ecoregion. They may improve a particular validation period by identifying historically high-risk places without representing dynamic ignition opportunity. Including them solely because most fires are human-caused would create a false sense that the human process has been modelled.

- A human spark is not sufficient for a recorded wildfire; weather, fuel and vegetation determine whether an ignition survives and spreads enough to be reported.
- Static infrastructure may be forecast-available, but it can act as a geographic memorisation feature and degrade temporal transfer.
- The empirical feature-selection criterion remains out-of-time generalisation, not thematic completeness.
- The omission should be stated as a limitation and addressed with a separate human-augmented challenger rather than by weakening the primary model.

9.3 Recommended future anthropogenic extensions

- Wildland-urban interface or building-density exposure.
- Distance to roads, campsites, recreation sites and developed land rather than aggregate ecoregion road kilometres.
- Land use, ownership and agricultural activity.
- Utility distribution-line exposure and weather-dependent line-failure risk, not only transmission density.
- Day of week, public holidays and seasonal recreation indicators.
- Cause-specific labels separating human and lightning fires, with lightning observations or forecasts for the natural-ignition model.

10. Leakage-safe mixed-data preprocessing and sampling

10.1 The categorical failure

US_L3NAME is a nominal feature with approximately thirteen ecoregion categories. The first script selected non-numeric columns and raised an intentional error. Converting categories to integers and applying ordinary SMOTE would be invalid because Euclidean interpolation would treat category order and distance as meaningful.

10.2 Corrected fold-local pipeline

1. Split the 2013-2020 development data by unique dates into rolling training and validation blocks.
2. Fit the numeric imputer only on the training rows of the current fold.
3. Scale numeric data only when required by distance-based imputation or sampling.
4. Impute US_L3NAME by training-fold mode and ordinal-encode it with a reserved unknown value.
5. Apply SMOTENC only to the transformed training fold, with the ecoregion column declared categorical.
6. Fit LightGBM and declare the transformed ecoregion index as categorical.
7. Evaluate untouched validation rows using AP, AP/prevalence, Brier score and log loss.

ADASYN and BorderlineSMOTE were removed from the primary mixed-data screen because there is no equivalent categorical-aware implementation in the selected workflow. They could be studied only under a different, explicitly labelled encoding experiment.

10.3 Class weights and sampling

Class weighting was treated as a separate imbalance method. Combining balanced class weights with SMOTENC would correct the class ratio twice and could strongly distort probabilities. The screening design therefore compared no correction, class weighting, random oversampling, SMOTENC and SMOTENC-Tomek as mutually exclusive branches.

11. Sampler-screen results

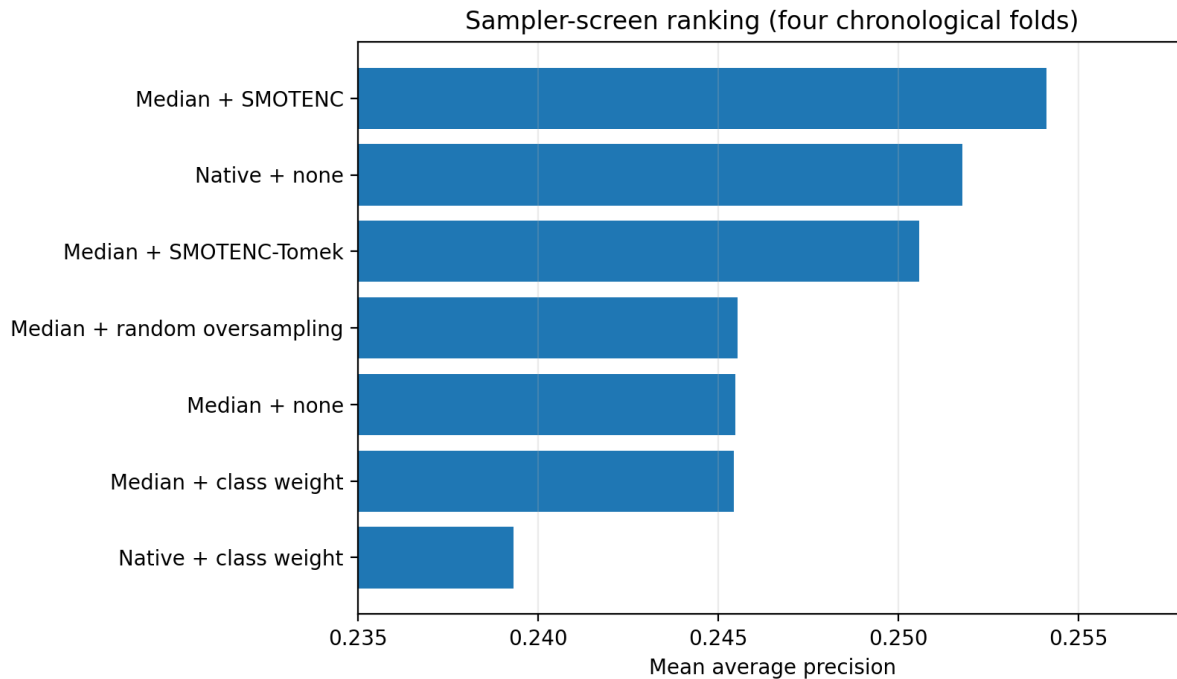


Figure 9. Mean AP in the sampler screen. The configuration name "SMOTE" in experiment logs denotes SMOTENC in the categorical-safe implementation.

Configuration	Mean AP	AP/prevalence	Brier	Log loss	Runtime (s)
Median + SMOTENC	0.2541	7.097	0.03616	0.14018	4.51
Native + none	0.2518	6.849	0.03328	0.15716	1.88 corrected
Median + SMOTENC-Tomek	0.2506	7.002	0.03636	0.14125	7.01
Median + random oversampling	0.2455	6.858	0.03638	0.14440	3.59
Median + none	0.2455	6.739	0.03340	0.15835	3.12
Median + class weight	0.2454	6.714	0.04683	0.16973	3.31
Native + class weight	0.2393	6.454	0.04702	0.17148	4.17

11.1 Fold-level reversal

Median + SMOTENC exceeded native AP in Folds 1 and 2 by approximately 0.017 and 0.036, but lost in Folds 3 and 4 by approximately 0.016 and 0.028. Brier was worse in three of four folds. The small aggregate AP advantage was therefore not stable through time and was driven largely by Fold 2.

Professional conclusion: native handling remained the provisional robust favourite; SMOTENC advanced as the main ranking challenger; SMOTENC-Tomek was deferred; class weighting and random oversampling were eliminated.

12. Imputer-screen results and finalist interpretation

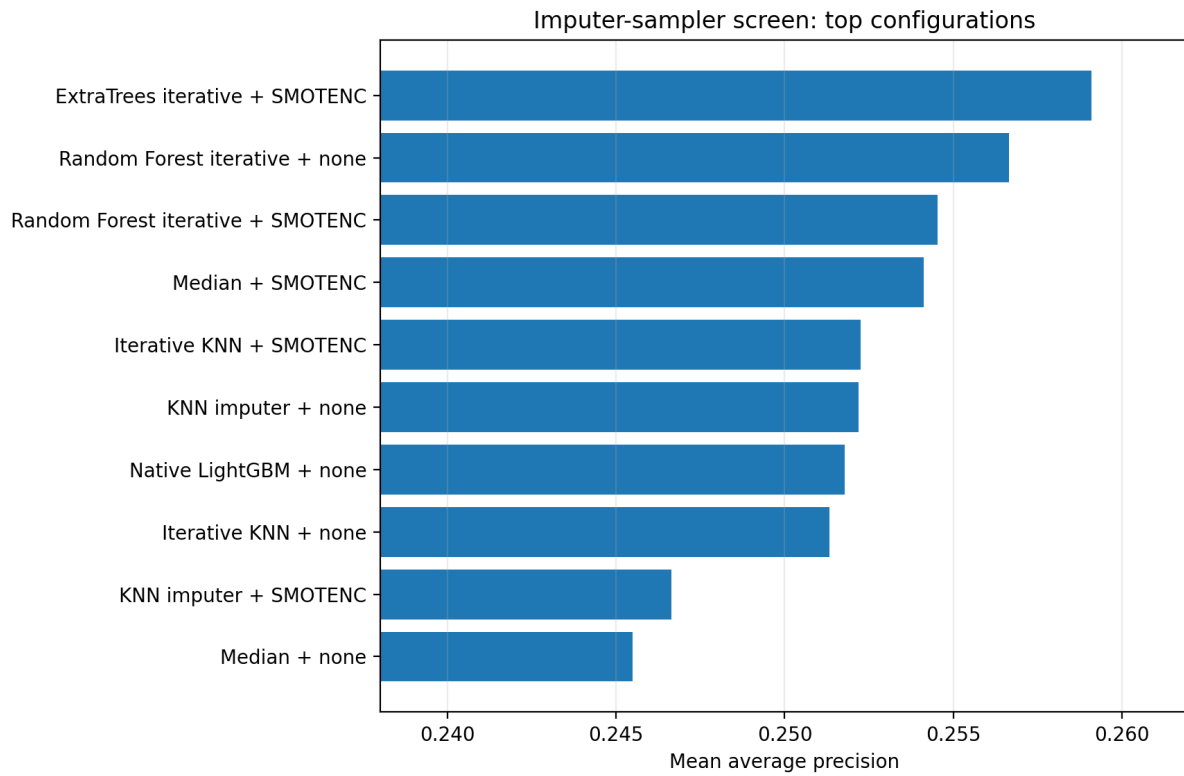


Figure 10. Top configurations in the imputer-sampler screen after correcting the duplicated native rows for descriptive statistics.

Configuration	Mean AP	AP SD	Brier	Log loss	Runtime
ExtraTrees iterative + SMOTENC	0.25910	0.09713	0.03552	0.13798	2,069 s
RF iterative + none	0.25665	0.11677	0.03315	0.15890	39,210 s
RF iterative + SMOTENC	0.25453	0.09651	0.03598	0.13937	6,474 s
Median + SMOTENC	0.25412	0.10048	0.03616	0.14018	4.56 s
Iterative KNN + SMOTENC	0.25226	0.10391	0.03607	0.14102	64.0 s
KNN imputer + none	0.25218	0.12002	0.03332	0.15957	207 s
Native + none	0.25177	0.11637	0.03327	0.15716	1.88 s
Bayesian iterative + SMOTENC	0.24460	0.10223	0.03460	0.13610	80.3 s
ExtraTrees iterative + none	0.24126	0.10689	0.03354	0.15956	2,470 s

12.1 ExtraTrees iterative + SMOTENC

This was the strongest mean-AP pipeline and the best SMOTENC imputer. Relative to median + SMOTENC, it improved AP in three folds, Brier in all four folds and log loss in all four folds. The mean AP gain was only about 0.005, however, while runtime increased from seconds to roughly 34.5 minutes. It is the performance finalist, not an automatic production winner.

12.2 Native LightGBM + no sampling

Native handling remained extremely competitive, had lower uncalibrated Brier than sampled pipelines, required no imputation model or synthetic observations, and performed better in the two latest folds. It is the mandatory robustness and engineering control.

12.3 Random Forest iterative + no sampling

RF iterative + none improved AP over native in three folds and essentially tied in the fourth, making it scientifically more credible than its mean alone suggested. The average gain was still only about 0.0049, log loss was slightly worse and total runtime exceeded ten hours. Fold 4 alone consumed almost ten hours while the corresponding sampled pipeline was much faster, which suggests a system or timing anomaly. The condition should be rerun before advancing.

12.4 Median + SMOTENC

Median + SMOTENC remained the efficiency benchmark: its AP was within approximately 0.005 of ExtraTrees while running hundreds of times faster. It is valuable for testing whether sophisticated imputation adds enough benefit to justify operational complexity.

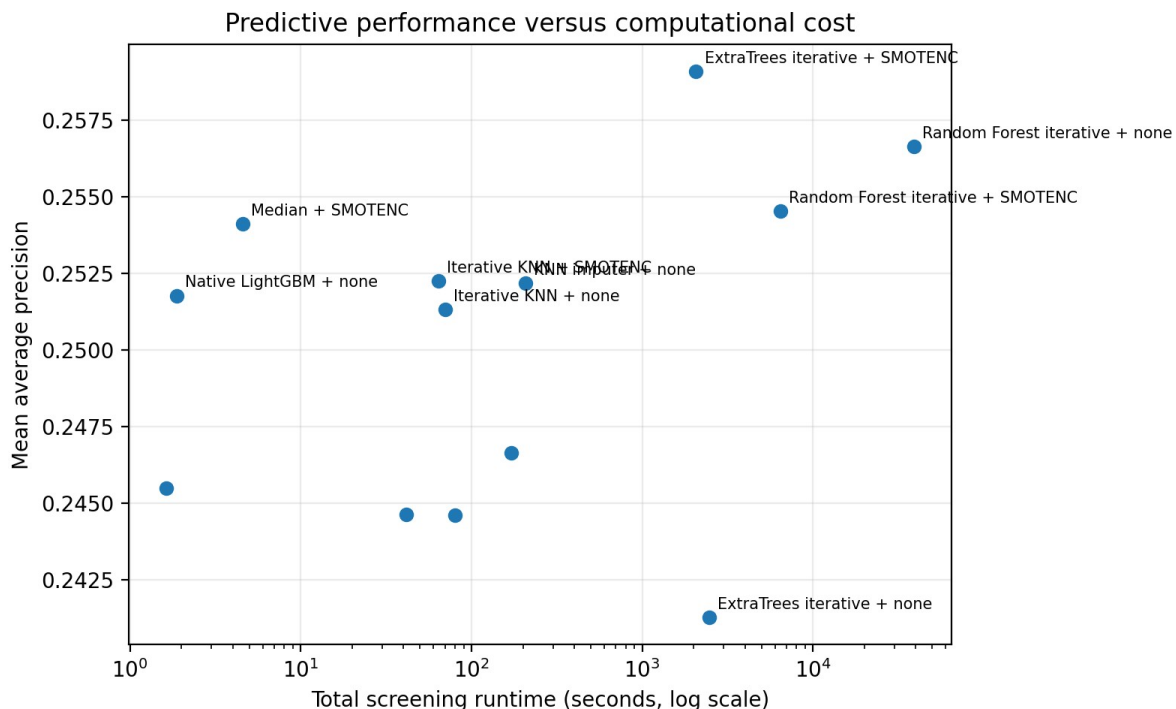


Figure 11. Runtime-performance trade-off. Several complex imputers provide only small AP changes relative to native or median pipelines.

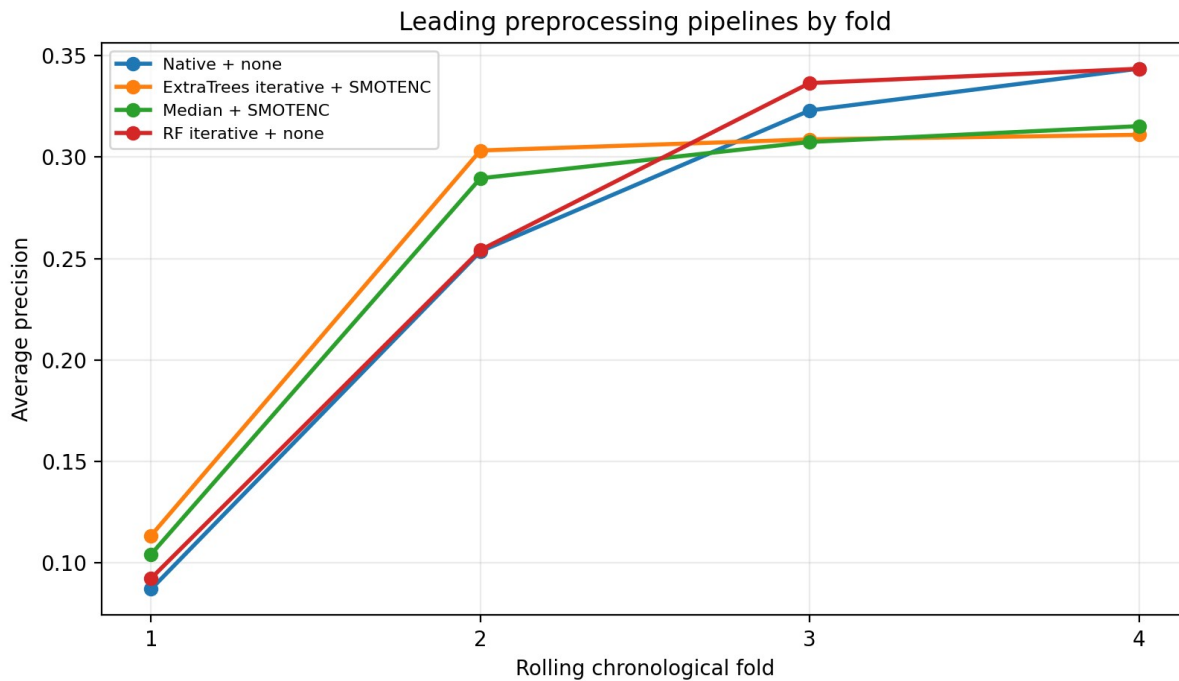


Figure 12. Leading preprocessing pipelines by rolling fold. Sampled pipelines lead early folds; native and no-sampling methods lead later folds.

12.5 Metric interpretation

SMOTENC pipelines generally had worse Brier scores but better log loss. This pattern is plausible: oversampling can raise probabilities for many negative cases, worsening squared error, while preventing extremely low probabilities on actual fire cases, improving log loss. Calibration may correct probability levels but will not normally create a major AP improvement. Final comparisons must therefore be repeated after sigmoid and isotonic calibration.

13. Error and resolution register

Issue	Consequence	Resolution / status
Test DataFrame passed as target	TypeError: unhashable DataFrame and accidental test involvement.	Correct function call; use training and validation only for diagnostics.
pd.qcut applied to DataFrame	Invalid binning object.	Pass the single feature Series.
Validation outside train bin range	NaN quantile assignments.	Use infinite outer bin edges.
AP computed from hard predictions in an early workflow	PR curve thresholds collapsed and AP was invalid.	Use predict_proba for AP, Brier and PR curves; hard predictions only for threshold metrics.
Row-based TimeSeriesSplit	Same date could enter train and validation through different ecoregions.	Split unique dates, then map dates back to row indices.
Numeric-only make_xy	US_L3NAME dropped and year could remain unintentionally.	Freeze an explicit feature list and process categorical predictors separately.
Ordinary SMOTE with ecoregion	Synthetic fractional category codes.	Use fold-local OrdinalEncoder plus SMOTENC.
Sampling with class_weight=balanced	Double correction and distorted probabilities.	Treat class weighting as a mutually exclusive imbalance method.
Every candidate tuned/calibrated/tested	Expensive, confounded selection and test leakage.	Use fixed model for screens; tune/calibrate only finalists; test once.
Baseline manually inserted	Reproducibility uncertainty.	Rerun baseline through identical experiment function and dataset version.
Duplicate native+none configuration rows	Means unchanged, but SD, runtime, bootstrap and paired comparisons distorted.	Validate configuration uniqueness and duplicate configuration-fold keys.
MissForest absent from results	Incomplete advertised screen.	Capture failure reason, repair or report exclusion explicitly.
RF Fold 4 runtime anomaly	Potentially misleading computational conclusion.	Rerun condition with system monitoring and pipeline caching.
Legacy "summer" script name	Documentation mismatch with full-year development data.	Rename scripts and artifacts to reflect current scope.
Test distributions already inspected	Holdout is not perfectly blinded.	Disclose and make no further test-driven decisions.

13.1 Recommended defensive checks

```

configs = stage_configurations()
assert len(configs) == len(set(configs)), 'Duplicate configurations'

keys = ['imputer', 'imbalance_method', 'fold']
assert not fold_results.duplicated(keys).any(), 'Duplicate fold rows'

assert set(train_dates).isdisjoint(set(validation_dates))
assert list(X.columns) == frozen_feature_list
assert 'fire_count' not in X and 'fire_occurred' not in X

```

14. Current research decision register

Component	Current choice	Alternative retained	Reason
Target/unit	Binary ecoregion-day occurrence	Cause-specific targets for future work	Matches current application and available labels.
Split	Chronological 2013-2020 / 2021-2022 / 2023-2025	Rolling unique-date folds inside training	Preserves future generalisation and supports development uncertainty.
Feature set	36-feature compact without transmission	37-feature compact with transmission	Best mean fold results and bootstrap-supported probability quality.
Categorical handling	Fold-local ordinal encoding; declare LightGBM categorical	Native categorical data frame in non-sampling pipeline	Required for SMOTENC and unknown categories.
Primary metric	Average precision	AP/prevalence context; Brier/log loss secondary	Rare-event ranking is primary; probability quality remains essential.
Sampler	Not yet frozen	None and SMOTENC finalists	Mean advantage is small and temporally unstable.
Imputer	Not yet frozen	Native, ExtraTrees iterative, median, conditional RF iterative	Different performance-runtime trade-offs.
Calibration	Pending	Sigmoid and isotonic	Must be selected on fixed validation after tuning.
Threshold	Pending	F2-based validation threshold	Application values recall, but threshold must not drive feature selection.
Second model family	XGBoost after LightGBM freeze	Same frozen workflow	Prevents repeated test adaptation and enables fair comparison.

15. Required next experiments and freeze protocol

15.1 Immediate data and code corrections

8. Remove the duplicate native+none configuration and regenerate all summaries from unique configuration-fold rows.
9. Record hashes or immutable filenames for the dataset and exact 36-feature list.
10. Capture MissForest failure messages. Repair only if the implementation supports fit on training and transform on future data; otherwise report exclusion.
11. Rerun RF iterative + none, especially Fold 4, to determine whether the ten-hour timing was reproducible.
12. Save out-of-fold probabilities for every finalist, including date, ecoregion, fold, y_true and probability.

15.2 Finalist uncertainty comparison

Run paired temporal block bootstrap comparisons using the same validation rows and block definition for:

- ExtraTrees iterative + SMOTENC minus native + none.
- ExtraTrees iterative + SMOTENC minus median + SMOTENC.
- RF iterative + none minus native + none.

Report AP, Brier and log-loss differences with bootstrap standard errors, percentile intervals and the probability of a favourable difference. The comparison must use row-level paired predictions, not four fold means.

15.3 Hyperparameter tuning

Tune no more than two or three pipelines. Use 45 RandomizedSearchCV iterations with average precision as refit metric, Brier and log loss as secondary scores, and the same unique-date rolling splits. Keep model n_jobs=1 inside parallel search to avoid oversubscription. Pipeline caching is strongly recommended for expensive imputers so they are not recomputed for every LightGBM parameter draw.

```
scoring = {
    'average_precision': 'average_precision',
    'brier': 'neg_brier_score',
    'log_loss': 'neg_log_loss',
}
refit = 'average_precision'
cv = unique_date_rolling_splits
```

15.4 Validation, calibration and threshold

13. Fit each tuned finalist on the original 2013-2020 training period.
14. Evaluate raw probabilities on 2021-2022.
15. Fit sigmoid and isotonic calibration using the frozen fitted pipeline and validation labels.
16. Compare calibrated AP, Brier, log loss and reliability plots. AP should remain nearly unchanged; Brier/log loss decide calibration quality.
17. Select an F2 threshold on validation only and report precision, recall, F1, F2, balanced accuracy and MCC as operating-point metrics.
18. Freeze the complete feature-preprocessing-model-calibration-threshold pipeline.

15.5 Final test and XGBoost

After the complete LightGBM pipeline is frozen, refit using the intended train-plus-validation protocol and evaluate the 2023-2025 test once. The XGBoost branch should then repeat the development workflow with the feature set and evaluation protocol fixed in advance. XGBoost must not be used as a reason to reopen LightGBM feature selection after seeing test performance.

16. Proposed methodological contribution for the new paper

The new paper can present a structured methodology for rare, nonstationary ecoregion-day wildfire occurrence prediction. The contribution is not a novel algorithm. It is a defensible model-development protocol that integrates operational feature availability, distribution diagnostics, temporally blocked ablation, prevalence context, paired uncertainty estimation and categorical-safe resampling.

16.1 Proposed methods sequence

19. Define forecast-time availability and remove direct leakage.
20. Audit duplicates, missingness, outliers, temporal drift and redundancy.
21. Use training-defined ECDF/quantile diagnostics to design targeted ablations.
22. Conduct feature-family and within-family ablations with a fixed model.
23. Confirm finalists across unique-date rolling folds and report AP relative to fold prevalence.
24. Use paired temporal block bootstrap to quantify uncertainty in finalist differences.
25. Freeze the primary feature set before screening imbalance and imputation methods.
26. Use fold-local mixed-data preprocessing and categorical-safe SMOTENC.
27. Screen methods with a fixed model, then tune only a small finalist set.
28. Calibrate and choose thresholds on a fixed validation period, followed by one out-of-time test.

16.2 Paper-ready feature-selection paragraph

Draft manuscript language

Candidate predictors were first audited for forecast-time availability, missingness, temporal drift and redundancy. Feature families were then evaluated using a fixed moderate-capacity LightGBM classifier so that changes in performance reflected feature inclusion rather than model retuning. Average precision was the primary metric, with Brier score and log loss used to assess probability quality. Leading feature specifications were subsequently compared across four rolling chronological folds constructed from unique dates. Because validation prevalence varied substantially between folds, AP was reported together with AP divided by prevalence. Final feature-set differences were quantified using a paired temporal block bootstrap on out-of-fold probabilities. This process selected a 36-feature compact specification that excluded unstable infrastructure proxies and reduced redundant observation-count and vegetation variables.

16.3 Paper-ready preprocessing paragraph

Draft manuscript language

After feature selection was frozen, imbalance and missing-data strategies were screened within the same rolling chronological folds. Numeric imputers were fitted only on each training fold. Ecoregion was imputed and ordinal-encoded within the training fold, with an explicit code for unseen categories. Synthetic oversampling used SMOTENC, with ecoregion declared categorical, to prevent interpolation of nominal category codes. Sampling was applied only to training-fold observations. A fixed LightGBM model was used during screening; hyperparameters were tuned only for the final preprocessing pipelines. Native LightGBM missing-value handling without resampling was retained as the principal low-complexity control.

17. Limitations and threats to validity

- The target combines human- and lightning-caused fires, which have different mechanisms. Cause-specific labels would improve scientific interpretation.
- The primary model contains no explicit dynamic human-activity variables. Ecoregion and month may proxy stable patterns but do not model ignition opportunity directly.
- Only four rolling development folds were used. This provides useful temporal stress tests but limited direct inference from fold-level standard deviations.
- Wilson intervals in the quantile plots ignore temporal and spatial dependence.
- The paired block-bootstrap conclusion depends on the selected block definition. Sensitivity to block length and possible ecoregion clustering should be documented.
- Infrastructure features are static or slowly varying at ecoregion scale and may memorise geography rather than transfer across space.
- The test period was descriptively inspected. Final performance remains valid as an out-of-time evaluation but should not be called a fully blinded holdout.
- The dataset changed between logs. Version control and a reproducible data-generation manifest are required before publication.
- MissForest was not successfully represented in the latest imputer results, and RF runtime measurements require confirmation.
- Calibration and final hyperparameter tuning have not yet been completed; current Brier/log-loss results are screening metrics, not final probability claims.
- Spatial transfer to unseen ecoregions is not evaluated because the same named ecoregions occur across periods. A future spatial holdout would answer a different generalisation question.

18. Overall professional conclusion

The project has moved from a broad collection of model experiments to a coherent research methodology. The strongest aspect is the deliberate separation of questions: first determine a defensible feature set, then evaluate imbalance and imputation strategies, then tune, calibrate and test. The chronological and paired analyses prevent a single favourable period from determining the conclusion.

The 36-feature compact model without transmission is sufficiently supported to freeze as the primary feature set. The absence of explicit human variables must be acknowledged, but it is more defensible than retaining unstable infrastructure proxies solely for thematic completeness. A separate human-augmented or cause-specific extension should address that limitation.

Among preprocessing candidates, ExtraTrees iterative + SMOTENC currently provides the strongest mean ranking, native LightGBM provides the strongest simplicity and recent-period robustness, and median + SMOTENC provides an exceptional efficiency benchmark. RF iterative + none is a conditional scientific finalist but requires a runtime rerun. The final winner should be chosen from paired out-of-fold uncertainty analysis and post-calibration probability quality, not from the current mean table alone.

Research-supervisor assessment

The 29 July-4 August work is strong enough to form the central methodological section of a serious paper, provided the remaining reproducibility issues are corrected and the final test is protected. The report supports a clear stopping rule: do not continue feature fishing; fix the experiment bookkeeping, compare only the finalists, tune and calibrate, then evaluate the out-of-time test once.

Appendix A. Exact frozen 36-feature primary set

Group	Features	Count
Dew point / humidity / dryness	dew_point_c_max dew_point_c_mean dew_point_c_min dew_point_c_missingRate sat_vp A_sat_vp RH_pct VPD_kpa HDWI_proxy	9
Wind	ew_wind_mean ns_wind_mean wind_speed_ms_max wind_speed_ms_mean wind_speed_ms_min wind_speed_ms_std wind_speed_ms_missingRate wind_dir_deg_missingRate wind_type_missingRate	9
Pressure / ceiling / visibility	sea_level_pressure_hPa_mean sea_level_pressure_hPa_missingRate ceiling_height_m_mean ceiling_height_m_missingRate vis_m_max vis_m_missingRate	6
Temperature	temp_c_max temp_c_mean temp_c_min temp_c_std DTR temp_c_missingRate temp_c_missingFlag_count	7
Vegetation	evi_mean evi_std	2
Temporal / categorical / completeness	month US_L3NAME row_weather_missing_rate	3

This list corresponds to the categorical-safe screening script. It contains no transmission, road, population, NDVI, raw year or observation-count variables. The list should be saved to an immutable JSON artifact and compared exactly before every experiment.

Appendix B. Feature-set rolling-fold results

Feature set	Fold	Prevalence	AP	AP/prev.	Brier	Log loss
Baseline all features	1	0.0130	0.0908	6.97	0.01278	0.08507
Baseline all features	2	0.0537	0.2459	4.58	0.05139	0.28302
Baseline all features	3	0.0342	0.2876	8.40	0.02864	0.11431
Baseline all features	4	0.0523	0.3092	5.91	0.04193	0.16298
Compact + transmission	1	0.0130	0.0820	6.29	0.01288	0.08296
Compact + transmission	2	0.0537	0.2426	4.51	0.05062	0.26633
Compact + transmission	3	0.0342	0.3233	9.44	0.02837	0.12180
Compact + transmission	4	0.0523	0.3399	6.50	0.04179	0.16406
Compact - transmission	1	0.0130	0.0915	7.02	0.01268	0.08098
Compact - transmission	2	0.0537	0.2535	4.72	0.05044	0.26110
Compact - transmission	3	0.0342	0.3230	9.43	0.02835	0.12195
Compact - transmission	4	0.0523	0.3436	6.57	0.04162	0.16286
Spatial keep road	1	0.0130	0.0938	7.19	0.01267	0.08495
Spatial keep road	2	0.0537	0.2491	4.63	0.05149	0.28231
Spatial keep road	3	0.0342	0.2860	8.35	0.02870	0.11525
Spatial keep road	4	0.0523	0.3311	6.33	0.04133	0.15978
Compact + roads	1	0.0130	0.0910	6.98	0.01272	0.08125
Compact + roads	2	0.0537	0.2504	4.66	0.05052	0.26801
Compact + roads	3	0.0342	0.2879	8.41	0.02934	0.12471
Compact + roads	4	0.0523	0.3378	6.46	0.04155	0.16256

Appendix C. Corrected imputer-screen summary

Imputer	Sampling	Mean AP	AP SD	AP/prev.	Brier	Log loss	Runtime s
ExtraTrees iterative	SMOTENC	0.259102	0.097129	7.326743	0.035516	0.137978	2,068.5
RF iterative	none	0.256646	0.116765	7.052487	0.033149	0.158902	39,210.4
RF iterative	SMOTENC	0.254529	0.096509	7.183229	0.035976	0.139365	6,473.9
Median	SMOTENC	0.254123	0.100484	7.097293	0.036161	0.140180	4.6
Iterative KNN	SMOTENC	0.252259	0.103912	6.959553	0.036071	0.141022	64.0
KNN imputer	none	0.252184	0.120025	6.808552	0.033318	0.159575	207.3
Native	none	0.251775	0.116366	6.848761	0.033275	0.157162	1.9
Iterative KNN	none	0.251324	0.117750	6.668470	0.033221	0.157687	70.3
KNN imputer	SMOTENC	0.246645	0.105943	6.711371	0.036206	0.140455	171.5
Median	none	0.245477	0.114383	6.739351	0.033402	0.158350	1.6
Bayesian iterative	none	0.244634	0.114546	6.715543	0.033353	0.160016	41.3
Bayesian iterative	SMOTENC	0.244603	0.102225	6.739537	0.034600	0.136105	80.3
ExtraTrees iterative	none	0.241263	0.106888	6.630226	0.033535	0.159562	2,470.0

Native+none duplicates were removed before computing the corrected standard deviation and runtime. MissForest is absent because no successful result was present in the supplied fold-results file.

Appendix D. Reproducibility checklist

- Immutable dataset file or hash and data-generation commit.
- Exact train, validation and test row counts and positive counts.
- Exact 36-feature JSON and categorical-feature list.
- Unique-date fold boundaries and row indices saved to disk.
- All preprocessing objects inside the fold-local pipeline.
- Sampler strategy, random state and categorical indices recorded.
- LightGBM parameters, package versions and CPU parallelism recorded.
- Raw fold-level predictions and metrics retained, not only summaries.
- Duplicate configuration and duplicate configuration-fold assertions.
- Failure log for every skipped configuration, including MissForest.
- Calibration data and threshold-selection data separated from tuning folds.
- Final test script disabled until a frozen-pipeline flag or artifact is present.

Appendix E. Source materials and references

E.1 Internal project materials

- Project log: 29th July 2026(2).txt - initial audit, counts, drift, family inspection and methodological notes.
- Project log: 1st August 2026(1).txt - updated counts, drift, ablations and rolling finalists.
- train_summer_model(1).py - legacy training, tuning, calibration and test-evaluation pipeline.
- screen_imputation_sampling(1).py - first numeric-only screening implementation and categorical failure.
- screen_imputation_sampling_categorical.py - corrected mixed-data pipeline with SMOTENC.
- imputer_screen_fold_results.csv - fold-level imputer and sampling results supplied on 4 August 2026.
- Conversation-supplied tables - fold prevalence, paired temporal block-bootstrap and sampler-screen summaries through 4 August 2026.
- Diagnostic figures supplied 31 July 2026 - ECDF and training-derived quantile-bin plots.

E.2 External references used for anthropogenic context

- Balch, J. K., Bradley, B. A., Abatzoglou, J. T., Nagy, R. C., Fusco, E. J., & Mahood, A. L. (2017). Human-started wildfires expand the fire niche across the United States. *Proceedings of the National Academy of Sciences*, 114(11), 2946-2951. doi:10.1073/pnas.1617394114.
- Mann, M. L., Batllori, E., Moritz, M. A., Waller, E. K., Berck, P., Flint, A. L., Flint, L. E., & Dolfi, E. (2016). Incorporating anthropogenic influences into fire probability models: Effects of human activity and climate change on fire activity in California. *PLOS ONE*, 11(4), e0153589. doi:10.1371/journal.pone.0153589.
- Keeley, J. E., & Syphard, A. D. (2018). Historical patterns of wildfire ignition sources in California ecosystems. *International Journal of Wildland Fire*, 27(12), 781-799. doi:10.1071/WF18026.
- Chen, B., & Jin, Y. (2022). Spatial patterns and drivers for wildfire ignitions in California. *Environmental Research Letters*, 17, 055004. doi:10.1088/1748-9326/ac60da.